

THE ERDŐS–GRAHAM IRREGULARITY PROBLEM FOR SUBSET SUMS

PRINCIPIA MATH

ABSTRACT. For $c > 0$, let $f_c(n)$ be the largest cardinality of a set $A \subseteq \{1, \dots, \lfloor cn \rfloor\}$ for which n is not a subset sum of A . We prove that $f_c(n)/n$ does not converge for every $0 < c < 1$. For $c \geq 1$ we have the exact formula $f_c(n) = \lfloor cn \rfloor - \lceil n/2 \rceil$, and hence $f_c(n)/n \rightarrow c - 1/2$. The only non-elementary input is a bounded zero-sum theorem of Alon.

For a finite set A of positive integers, write

$$\Sigma(A) = \left\{ \sum_{a \in B} a : B \subseteq A \right\}, \quad F(M, n) = \max\{|A| : A \subseteq [1, M], n \notin \Sigma(A)\}.$$

Thus $f_c(n) = F(\lfloor cn \rfloor, n)$. Erdős and Graham asked whether this extremal quantity depends irregularly on n [2, p. 59]. We prove the following complete dichotomy.

Theorem 1. *Let $c > 0$.*

- (1) *If $0 < c < 1$, then $f_c(n)/n$ does not converge.*
- (2) *If $c \geq 1$, then*

$$f_c(n) = \lfloor cn \rfloor - \left\lceil \frac{n}{2} \right\rceil \quad (n \geq 1),$$

so $f_c(n)/n \rightarrow c - 1/2$.

We use the following theorem of Alon [1, Theorem 1.1]: for every integer $k \geq 2$ and every $\varepsilon > 0$, there is N_0 such that, whenever $N \geq N_0$ and $X \subseteq \mathbb{Z}/N\mathbb{Z}$ satisfies

$$|X| > \left(\frac{1}{k} + \varepsilon \right) N,$$

there is a nonempty $Y \subseteq X$ with $|Y| \leq k$ and $\sum_{y \in Y} y = 0$ in $\mathbb{Z}/N\mathbb{Z}$.

The next consequence is the only additive input needed below.

Proposition 2. *Fix $\rho \geq 2$ and $\delta > 0$. For all sufficiently large E , the following holds uniformly for every even integer τ with*

$$2E \leq \tau \leq \rho E.$$

If $A \subseteq [1, E]$ and $\tau \notin \Sigma(A)$, then, with

$$K = \left\lfloor \frac{3\tau}{2E} \right\rfloor,$$

we have

$$|A| \leq \frac{\tau}{2K} + \delta E.$$

Proof. Put $q = \tau/2$. Then $E \leq q \leq \rho E/2$, while $3 \leq K \leq 3\rho/2$; hence K ranges over a fixed finite set. We may therefore use Alon's theorem uniformly in K , with $\varepsilon = \delta/\rho$, and take E so large that also $K < \delta E/2$.

Suppose that $|A| > q/K + \delta E$. Reduction modulo q is injective on $[1, E]$, and

$$\frac{q}{K} + \frac{\delta q}{\rho} \leq \frac{q}{K} + \frac{\delta E}{2} < |A|.$$

Alon's theorem gives a nonempty $B_1 \subseteq A$ with $|B_1| \leq K$ and $q \mid \sum_{b \in B_1} b$. Moreover

$$0 < \sum_{b \in B_1} b \leq |B_1|E \leq KE \leq 3q.$$

The last inequality is strict. Indeed, equality throughout would force $|B_1| = K \geq 3$ and every element of B_1 to equal E , which is impossible. Thus

$$\sum_{b \in B_1} b \in \{q, 2q\}.$$

If this sum is $2q = \tau$, we are done. Otherwise it is q . After deleting B_1 we still have

$$|A \setminus B_1| > \frac{q}{K} + \delta E - K > \frac{q}{K} + \frac{\delta E}{2} \geq \left(\frac{1}{K} + \frac{\delta}{\rho} \right) q.$$

A second application of Alon's theorem gives a nonempty $B_2 \subseteq A \setminus B_1$ whose ordinary sum is again q or $2q$. In the latter case B_2 sums to τ ; in the former case $B_1 \cup B_2$ does. Hence every set larger than the asserted bound contains a subset summing to τ . $\square$

Proof of Theorem 1. First suppose $0 < c < 1$, and assume for contradiction that

$$\frac{f_c(n)}{n} \longrightarrow L.$$

For odd n , the even elements of $[1, \lfloor cn \rfloor]$ have no subset sum equal to n . Their cardinality is $\lfloor \lfloor cn \rfloor / 2 \rfloor$, so passage to the odd subsequence gives

$$L \geq \frac{c}{2}. \tag{1}$$

Suppose first that $c \leq 1/2$, and put

$$J = \left\lfloor \frac{3}{2c} \right\rfloor.$$

Since $\lfloor x \rfloor > x - 1$ and $c \leq 1/2$, we have $J > 1/c$. Choose $\delta > 0$ so that

$$\beta := \frac{1}{2J} + \delta c < \frac{c}{2}.$$

For sufficiently large even n , set $M = \lfloor cn \rfloor$. Then $2M \leq n \leq (2/c)M$, and Proposition 2, applied with $E = M$ and $\tau = n$, gives

$$f_c(n) \leq \frac{n}{2K} + \delta M, \quad K = \left\lfloor \frac{3n}{2M} \right\rfloor \geq J.$$

Consequently $f_c(n)/n \leq \beta$ for all sufficiently large even n , and therefore $L \leq \beta < c/2$, contradicting (1).

Now suppose $1/2 < c < 1$. Write $d = 1 - c$ and

$$J = \left\lfloor \frac{3}{2d} \right\rfloor.$$

Since $d < 1/2$, the same calculation gives $J > 1/d$. Set

$$g = \frac{d}{2} - \frac{1}{2J} > 0, \quad \delta = \frac{g}{3d}, \quad \eta = \frac{g}{3}.$$

Then

$$c - \frac{1}{2} + \frac{1}{2J} + \delta d + \eta = \frac{c}{2} - \frac{g}{3} < \frac{c}{2}. \tag{2}$$

For a sufficiently large even n , put $M = \lfloor cn \rfloor$ and $D = n - M$. If $A \subseteq [1, M]$ avoids n , split

$$A_0 = A \cap [1, D - 1], \quad A_1 = A \cap [D, M].$$

The interval $[D, M]$ is invariant under $x \mapsto n - x$. Since A cannot contain both members of a two-element orbit,

$$|A_1| \leq \left\lfloor \frac{M - D + 2}{2} \right\rfloor \leq \left(c - \frac{1}{2}\right)n + 1.$$

Since $dn \leq D < dn + 1$, for all sufficiently large n we have $D \leq M$, $2(D - 1) \leq n \leq (2/d)(D - 1)$, and $D - 1 \leq dn$. Apply Proposition 2 to A_0 with $E = D - 1$, $\tau = n$, and $\rho = 2/d$; it gives

$$|A_0| \leq \frac{n}{2K} + \delta(D - 1), \quad K = \left\lfloor \frac{3n}{2(D - 1)} \right\rfloor \geq J.$$

Since $D - 1 \leq dn$, we obtain

$$\frac{|A|}{n} \leq c - \frac{1}{2} + \frac{1}{2J} + \delta d + \frac{1}{n}.$$

Taking n large enough that $1/n \leq \eta$, (2) again gives an eventual upper bound strictly below $c/2$ on the even subsequence, contradicting (1). This proves part (1).

For part (2), let $1 \leq n \leq M$. An avoiding set cannot contain n . On $[1, n - 1]$, the map

$$x \mapsto \min\{x, n - x\}$$

is injective on every avoiding set, since two distinct elements with the same image sum to n . Hence an avoiding subset of $[1, M]$ has at most

$$\left\lfloor \frac{n}{2} \right\rfloor + (M - n) = M - \left\lceil \frac{n}{2} \right\rceil$$

elements. Equality is attained by

$$\left[\left\lceil \frac{n}{2} \right\rceil, n - 1 \right] \cup [n + 1, M],$$

which plainly has no subset sum equal to n . Thus

$$F(M, n) = M - \left\lceil \frac{n}{2} \right\rceil.$$

Taking $M = \lfloor cn \rfloor$ proves part (2). □

REFERENCES

- [1] N. Alon, *Subset sums*, J. Number Theory **27** (1987), 196–205.
- [2] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique, no. 28, Geneva, 1980.